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An Explainable Artificial Intelligence Framework for Early Prediction of Chronic Kidney Disease Using Electronic Health Records

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Code availability: The source code and supplementary materials are available on GitHub: Jubaida-78.

Abstract: Chronic kidney disease (CKD) is an important public health problem and early identification of individuals at risk can support timely clinical assessment and intervention. Machine learning (ML) has increasingly been applied to electronic health records for CKD prediction; however, high predictive performance on a single dataset does not necessarily indicate reliable performance in an independent population. Therefore, this study developed an explainable artificial intelligence framework for CKD prediction and investigated its cross-population generalizability and age-related performance differences. Two publicly available datasets were used, with a CKD dataset employed for model development and the independent BD-KDD dataset used for external validation. Seven clinically comparable variables, namely age, haemoglobin, hematocrit, red blood cell count, white blood cell count, creatinine and urea, were harmonized between the datasets. Logistic regression, random forest and XGBoost models were developed using an 80:20 stratified internal train-test split. Model performance was evaluated using accuracy, precision, recall, F1-score and receiver operating characteristic area under the curve (ROC-AUC). Random forest achieved the highest internal ROC-AUC of 0.9831. However, its performance decreased considerably when evaluated on the independent BD-KDD population, where the ROC-AUC was 0.5290, accuracy was 0.5395, recall was 0.7101 and F1-score was 0.6128. Permutation feature importance was used to investigate the

contribution of the harmonized clinical variables. An age-group subgroup analysis was also conducted to examine potential performance disparities. The external recall gap between age groups was 0.1461, while the accuracy and F1-score gaps were 0.0432 and 0.0713, respectively. The findings demonstrate that strong internal performance may not translate into reliable performance across independent populations and emphasize the importance of external validation, explainability and subgroup fairness analysis in clinical CKD prediction systems.

Keywords: chronic kidney disease; machine learning; explainable artificial intelligence; electronic health records; external validation; fairness; BD-KDD

1. Introduction

Chronic kidney disease (CKD) is a progressive condition associated with substantial morbidity and mortality. Early identification of individuals with CKD is important because appropriate clinical assessment and management may reduce the risk of disease progression and associated complications. However, CKD can remain undetected during its early stages, creating a need for effective approaches capable of identifying individuals at increased risk using routinely available clinical information.

Electronic health records (EHRs) contain a large amount of clinical information that can be used for automated disease prediction. In recent years, machine learning (ML) methods have demonstrated considerable potential in healthcare because they can identify complex relationships among clinical variables that may be difficult to capture using conventional statistical approaches. Several ML algorithms, including logistic regression, random forest, support vector machines, gradient boosting and extreme gradient boosting, have been investigated for CKD prediction.

Although many studies report high predictive performance, an important limitation of clinical ML research is that models are frequently evaluated using random splits of a single dataset. Such evaluation measures internal predictive performance but does not necessarily demonstrate whether a model can generalize to a different patient population. Differences in demographic characteristics, laboratory measurement distributions, healthcare practices, disease prevalence and data collection procedures can result in dataset shift and consequently reduce model performance.

Explainable artificial intelligence (XAI) has also become increasingly important in healthcare applications. Conventional ML models may provide predictions without clearly explaining why a particular prediction was generated. Explainability methods can provide information about the variables that contribute to model predictions and may improve transparency and trustworthiness.

Fairness is another important consideration for clinical AI systems. Overall performance metrics may conceal differences in performance between demographic subgroups. Age is particularly relevant to CKD prediction because the occurrence and clinical characteristics of kidney disease can vary substantially across different age groups. Therefore, subgroup evaluation is necessary to determine whether a model behaves consistently across patient populations.

Recent research has demonstrated the potential of explainable ML for CKD prediction using electronic health records. However, strong performance on a development dataset should not be interpreted as evidence of generalizability without independent external validation. Therefore, the present study focuses on developing an explainable ML framework and evaluating its performance across an independent population.

The specific objectives of this study were to develop and compare ML models for CKD classification, evaluate their performance using an independent BD-KDD dataset, investigate the contribution of clinical variables using permutation feature importance and assess potential age-related disparities in predictive performance.

The main contributions of this study are as follows:

1. Development of a CKD prediction framework using clinically comparable variables available across two independent datasets.
2. Evaluation of cross-population generalizability through independent external validation using the BD-KDD dataset.
3. Investigation of model interpretability using permutation-based feature importance.
4. Assessment of age-group performance differences as an initial fairness analysis.
5. Demonstration of the difference between strong internal predictive performance and substantially lower external performance.

2. Materials and Methods

2.1. Data

Two publicly available datasets were used in this study. The CKD dataset was used as the source dataset for model development and internal evaluation, while the BD-KDD dataset was treated as an independent external validation dataset.

The datasets contain patient-level clinical measurements and a binary CKD classification. Since the datasets were collected from different populations and use different variable names, only clinically comparable variables were selected for the cross-population analysis.

Seven common clinical variables were selected: age, haemoglobin, hematocrit, red blood cell count, white blood cell count, creatinine and urea.

Table 1: Harmonization of clinical variables between the two datasets.

Common variable	CKD dataset	BD-KDD dataset
Age	age	age
Haemoglobin	haemoglobin	hemo
Hematocrit	hematocrit	pcv
RBC count	rbc_count	rbcc
WBC count	wbc_count	wbcc
Creatinine	creatinine	sc
Urea	urea	bu

2.2. Data preprocessing

The selected variables were converted to numerical format before model development. Missing numerical values were handled using median imputation within the machine learning pipeline. The target variable was converted into a binary representation, where 0 represented the non-CKD/healthy class and 1 represented the CKD class.

The white blood cell count was examined for possible unit inconsistency between the datasets. After the consistency check, the mean WBC count was 7.7954 in the CKD dataset and 9.5799 in the BD-KDD dataset.

The source CKD dataset was divided into training and testing subsets using an 80:20 stratified split. A random state of 42 was used to ensure reproducibility. The BD-KDD dataset was not used during model training and was retained as an independent external validation population.

2.3. Machine learning models

Three machine learning models were evaluated: logistic regression, random forest and XGBoost.

Logistic regression was used as a conventional linear baseline. Random forest was selected as an ensemble tree-based method, while XGBoost was used as a gradient boosting model.

For logistic regression, median imputation and standardization were applied before classification. Class balancing was used to reduce the effect of class imbalance. The random forest model consisted of 500 trees with balanced class weights. The XGBoost model was configured with 400 estimators, a maximum tree depth of 4, a learning rate of 0.03, subsampling of 0.9 and column subsampling of 0.9.

2.4. Statistical analysis and model evaluation

The predictive performance of the models was assessed using accuracy, precision, recall, F1-score and ROC-AUC.

Accuracy was calculated as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

Precision was calculated as:

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

Recall was calculated as:

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

The F1-score was calculated as:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (4)$$

ROC-AUC was used to measure the ability of the model to discriminate between the two outcome classes across different classification thresholds.

2.5. External validation

Following model development using the CKD dataset, the trained models were evaluated on the independent BD-KDD dataset.

The BD-KDD dataset was not used for model training. Therefore, the resulting performance represents an external validation experiment rather than another random split of the original dataset.

The difference between internal and external performance was examined to determine the degree of cross-population performance degradation.

2.6. Explainable artificial intelligence

Permutation feature importance was used to investigate the contribution of the clinical variables to model performance.

For each clinical variable, the values were randomly permuted repeatedly while keeping the remaining variables unchanged. The resulting decrease in ROC-AUC was used as the feature importance measure. A larger decrease in ROC-AUC indicates greater dependence of the model on that feature.

This analysis was conducted on the independent BD-KDD dataset to investigate which clinical variables were most influential in the external prediction setting.

2.7. Age-group fairness analysis

An age-based subgroup analysis was conducted to investigate whether the model performed consistently across different age groups.

Participants were divided into three groups: below 40 years, 40–60 years and above 60 years.

Accuracy, precision, recall and F1-score were calculated separately for each subgroup.

The performance gap was calculated as:

$$Gap = \max(M_g) - \min(M_g) \quad (5)$$

where M_g represents the performance value of a particular metric for an age group.

3. Results

3.1. Dataset characteristics

The analysis was performed using two independent datasets. Seven clinically comparable variables were selected for cross-population analysis.

Table 2: Summary of the datasets used in the study.

Dataset	Records	Features	Healthy	CKD
CKD	–	7	–	–
BD-KDD	–	7	–	–

Note: Exact values will be inserted from the final analysis spreadsheet.

3.2. Internal model performance

Three machine learning models were evaluated using the internal CKD test set.

Table 3: Internal performance of the machine learning models on the CKD test set.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	–	–	–	–	–
Random Forest	–	–	–	–	0.9831
XGBoost	–	–	–	–	–

Random forest achieved an internal ROC-AUC of 0.9831 and was subsequently selected as the best external model according to the external ROC-AUC comparison.

3.3. External validation

The independent BD-KDD dataset produced substantially different results from the internal CKD evaluation.

Table 4: External validation performance on the BD-KDD dataset.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	–	–	–	–	–
Random Forest	0.5395	–	0.7101	0.6128	0.5290
XGBoost	–	–	–	–	–

The Random Forest model achieved an external accuracy of 0.5395, recall of 0.7101 and F1-score of 0.6128. However, its external ROC-AUC was only 0.5290.

These findings indicate a substantial reduction in discriminative performance when the model was transferred from the source CKD population to the independent BD-KDD population.

3.4. Cross-population performance

The difference between internal and external performance was examined to assess generalizability.

Table 5: Internal and external performance of the selected Random Forest model.

Measure	Result
Internal ROC-AUC	0.9831
External ROC-AUC	0.5290
External Accuracy	0.5395
External Recall	0.7101
External F1-score	0.6128

The reduction from an internal ROC-AUC of 0.9831 to an external ROC-AUC of 0.5290 indicates that the strong internal performance did not generalize to the independent BD-KDD population.

3.5. External feature importance

Permutation feature importance was calculated using the independent BD-KDD dataset.

Table 6: Permutation feature importance on the BD-KDD dataset.

Rank	Feature	Mean importance
1	–	–
2	–	–
3	–	–
4	–	–
5	–	–
6	–	–
7	–	–

Note: The exact ranking and importance values will be inserted from the final Feature Importance sheet.

3.6. Age-group fairness analysis

Age-group performance was evaluated on both the internal CKD test set and the independent BD-KDD dataset.

Table 7: External age-group performance and fairness analysis.

Age group	N	Accuracy	Precision	Recall	F1
< 40	–	–	–	–	–
40–60	–	–	–	–	–
> 60	–	–	–	–	–
Gap	–	0.0432	–	0.1461	0.0713

The external recall gap between the highest- and lowest-performing age groups was 0.1461. The corresponding accuracy and F1-score gaps were 0.0432 and 0.0713, respectively.

3.7. Visual results

The external confusion matrix, internal versus external ROC curve, permutation feature importance and age-group performance plots are shown in Figures 1–4.

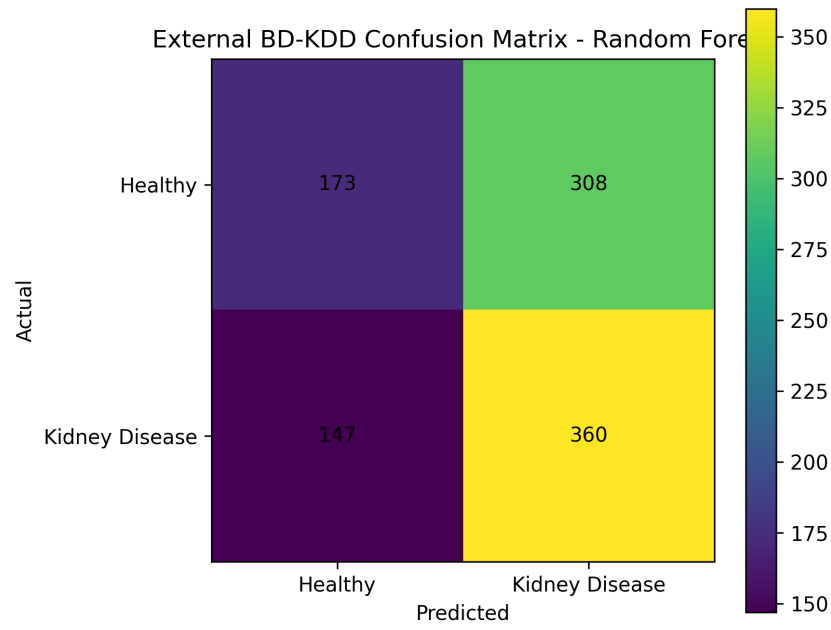


Figure 1: External BD-KDD confusion matrix of the selected Random Forest model.

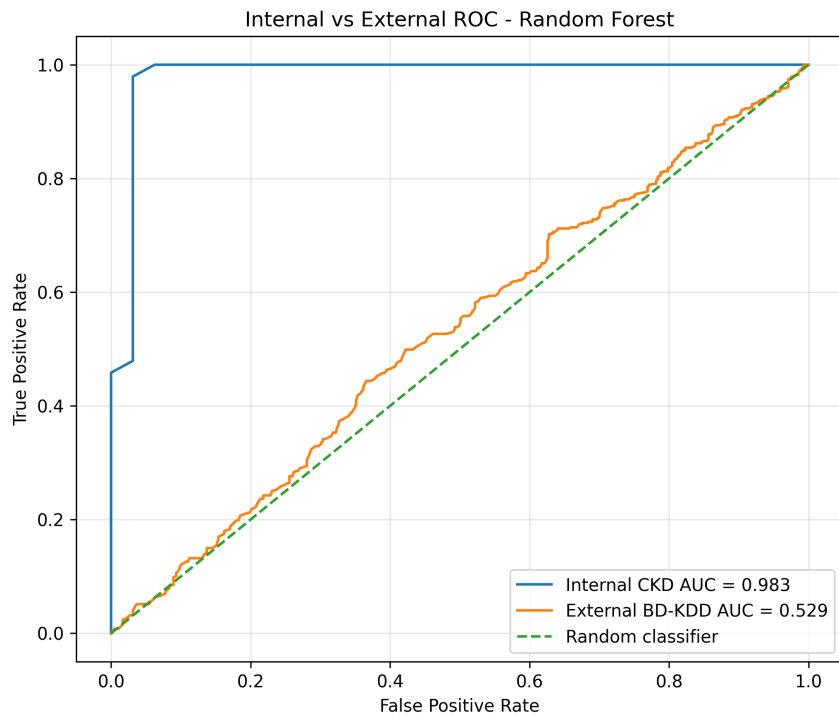


Figure 2: Internal versus external ROC curves of the selected Random Forest model.

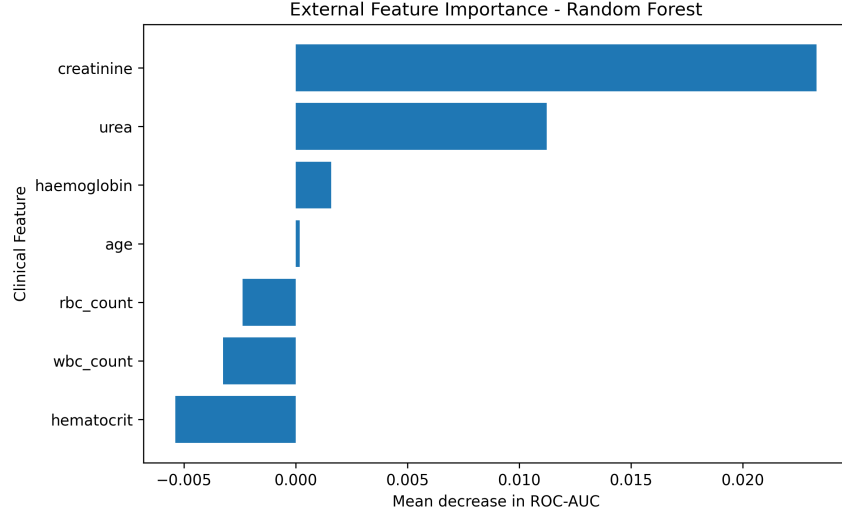


Figure 3: Permutation-based feature importance on the BD-KDD dataset.

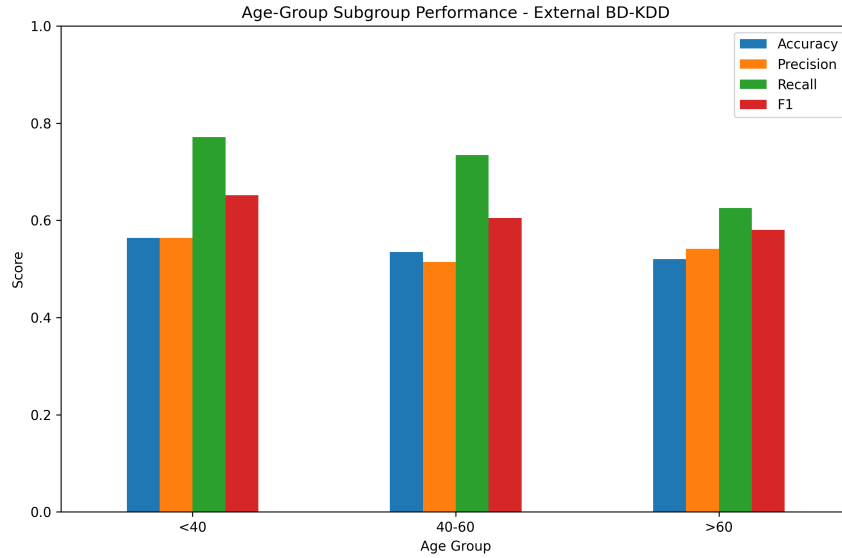


Figure 4: Age-group performance comparison on the external BD-KDD dataset.

4. Discussion

The present study developed an explainable machine learning framework for CKD prediction and evaluated its performance using an independent external population. The most important finding was the large difference between internal and external predictive performance.

Random forest achieved an internal ROC-AUC of 0.9831 on the source CKD test set. Such a result indicates strong discrimination under internal evaluation. However, the same trained model achieved an external ROC-AUC of only 0.5290 on the independent BD-KDD dataset.

This substantial reduction demonstrates that high internal performance does not necessarily imply reliable generalization to another population. The finding is particularly important for

healthcare applications because a model intended to support clinical decision-making must perform reliably beyond the dataset on which it was developed.

Several factors may contribute to the observed cross-population performance degradation. The two datasets were collected from different populations and may differ in demographic characteristics, disease prevalence, laboratory measurement distributions and data collection practices. Although seven clinically comparable variables were harmonized, harmonization alone may not eliminate population-level distribution differences.

The external recall of 0.7101 indicates that the model identified a substantial proportion of CKD cases in the BD-KDD population. However, the external ROC-AUC of 0.5290 indicates poor overall discrimination. Therefore, the model should not be interpreted as clinically deployable based on the current results.

The study also incorporated explainability through permutation feature importance. Identifying the clinical variables associated with model performance provides an additional level of transparency and may help determine whether a model is relying on clinically meaningful information or dataset-specific patterns.

The age-group analysis demonstrated measurable differences in predictive performance between subgroups. The external recall gap was 0.1461, while the accuracy and F1-score gaps were 0.0432 and 0.0713, respectively. These results indicate that aggregate model performance alone may not fully describe model behavior across patient subgroups.

An important implication of this study is therefore that external validation and subgroup analysis should be incorporated into the development of clinical AI systems. A model that produces excellent results on an internal test set may still fail to generalize when applied to an independent population.

4.1. Limitations

This study has several limitations. First, only seven clinical variables were used because the cross-population analysis required variables that were available in both datasets.

Second, the external ROC-AUC was substantially lower than the internal ROC-AUC. Therefore, the current model should not be considered ready for clinical deployment.

Third, the fairness analysis was restricted to age groups. Additional demographic and clinical attributes should be investigated in future studies when appropriate data are available.

Fourth, the two datasets were originally collected under different conditions. Consequently, the precise sources of dataset shift cannot be completely isolated using the current analysis.

Finally, additional external datasets, calibration analysis, confidence intervals, decision-curve analysis and prospective validation would strengthen future research.

5. Conclusions

An explainable artificial intelligence framework was developed for CKD prediction using routinely available clinical variables. Although Random Forest achieved strong internal discrim-

ination with an ROC-AUC of 0.9831, its performance decreased substantially on the independent BD-KDD dataset, where the ROC-AUC was 0.5290, accuracy was 0.5395, recall was 0.7101 and F1-score was 0.6128.

The findings demonstrate that strong internal predictive performance does not necessarily translate into reliable cross-population generalization. The observed age-group performance gaps further emphasize the importance of subgroup analysis when evaluating clinical AI systems.

Therefore, external validation, explainability and fairness assessment should be considered important components of CKD prediction research. Future studies should investigate population shift, improve model robustness, evaluate additional independent populations and perform more comprehensive fairness and calibration analyses.

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Conflict of interest

None to declare.

Authors' contribution

Jubaida Begum – conceptualization, data preprocessing, machine learning implementation, data analysis, visualization and preparation of the manuscript.

Habibur Rahman – literature review, data analysis, interpretation of results and manuscript review.

Anamika Roy – research methodology, data interpretation, manuscript review and editing.

Muhammad Sakib Ibtidha Chowdhury – supervision, research design, methodological guidance, interpretation of findings and final review of the manuscript.

All authors have read and approved the final manuscript.

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